

CELL: THE BUILDING BLOCK OF LIFE

Class 8 Science • NCERT Chapter 2 • Quick + Exam-Oriented Notes

CORE IDEA

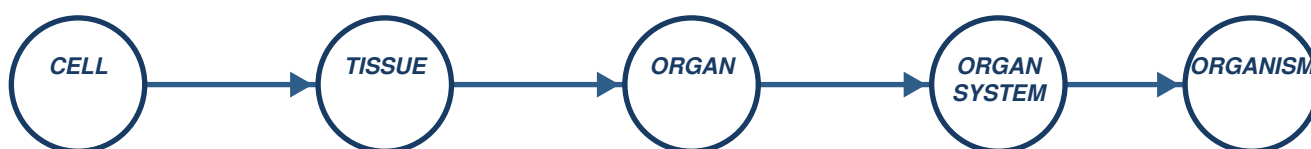
A cell is the basic unit of life. Every living organism is made of one or more cells.

1. What is a Cell?

A cell is the smallest basic unit that makes up a living organism and performs the essential functions needed for life.

- Unicellular organisms: made of only one cell. Examples: bacteria and yeast.
- Multicellular organisms: made of many cells. Examples: plants, fish, birds and humans.

Organisation in Multicellular Organisms



Cell → Tissue → Organ → Organ system → Organism

- A group of similar cells performing similar functions forms a tissue.
- Different tissues are organised to form an organ.
- Several organs work together to form an organ system.

REMEMBER

One cell = Unicellular | Many cells = Multicellular

2. How to Study Cells?

Most cells are too small to be seen clearly with the naked eye. Scientists therefore use microscopes to magnify and study them.

Limit of Resolution

The limit of resolution is the ability to see two very close points as two separate points. The human eye can distinguish two points about 0.1 mm apart.

EXAM FACT

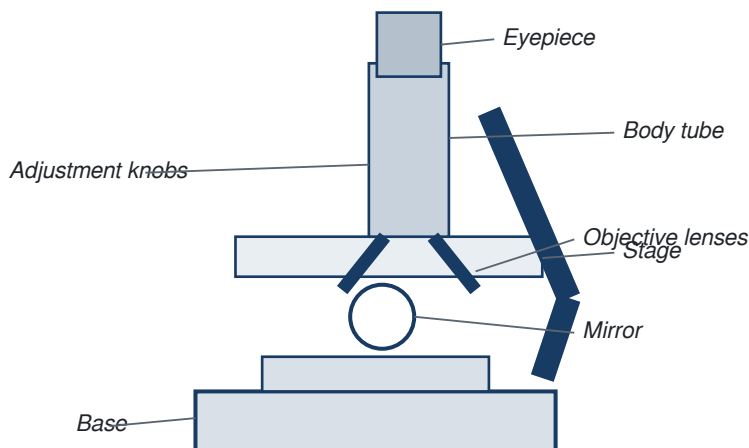
Human eye limit of resolution ≈ 0.1 mm.

3. Magnification & Microscopes

Magnification means making an object appear larger than its actual size. A microscope uses lenses to magnify tiny objects.

Why do we need a microscope?

- Most cells are much smaller than what the naked eye can clearly resolve.
- A microscope magnifies the object so that its structure can be observed.
- Different objective lenses can provide different magnifications, such as 10X and 40X.



Important Parts of a Light Microscope

Part	Main role
Eyepiece	The lens through which we look.
Objective lens	Magnifies the specimen.
Stage	Supports the slide/specimen.
Mirror	Helps direct light towards the specimen.
Adjustment knobs	Used for focusing the image.
Base	Provides support to the microscope.

QUICK REMEMBER

Magnification = making an object appear larger.
Common objective lenses shown in the chapter: 10X and 40X.

4. Robert Hooke & the Discovery of Cells

In 1665, Robert Hooke examined a thin slice of cork using a self-designed microscope.

- He observed many small, box-like compartments in the cork.
- He named these compartments “cells”.

MEMORY LINE

ROBERT HOOKE → 1665 → CORK → “CELLS”

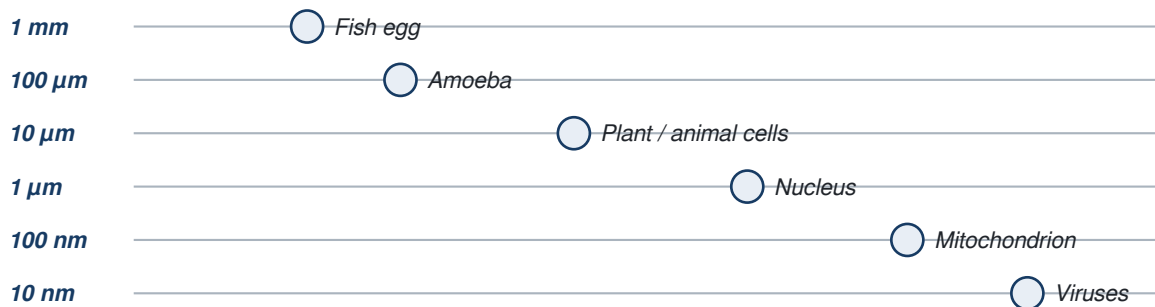
5. Light Microscope vs Electron Microscope

Light Microscope	Electron Microscope
Uses light to form the image.	Uses a beam of electrons.
Commonly used in school laboratories.	Used to reveal much finer details of cell structures.
Suitable for observing cells with light.	Can reveal structures at the nanometre scale.

6. Size of Microscopic Objects

Cells and other tiny structures are described using small units such as millimetres (mm), micrometres (μm) and nanometres (nm).

Approximate size range of microscopic objects



UNIT CONVERSION

1 millimetre (mm) = 1000 micrometres (μm)

7. Estimating the Size of a Cell

The size of a cell can be estimated by measuring the diameter of the visible field of view and counting how many cells fit along that diameter.

Formula ☆

Size of one cell = Diameter of visible field ÷ Number of cells along the diameter

WORKED EXAMPLE

If the visible field diameter is 5000 μm and 25 cells fit along the diameter:

Cell size = $5000 \div 25 = 200 \mu\text{m}$

Steps in the NCERT Activity

- Place a transparent ruler with millimetre markings on the microscope stage.
- Focus on the ruler and observe the diameter of the visible field.
- Convert the diameter into micrometres (μm).
- Replace the ruler with an onion peel slide.
- Count the number of cells along the diameter of the field of view.
- Use the formula to estimate the actual size of one cell.

8. Chapter Quick Revision

Concept	Must remember
Basic unit of life	Cell
One-cell organism	Unicellular
Many-cell organism	Multicellular
Human eye resolution	About 0.1 mm
Robert Hooke	Observed cork cells in 1665
Magnification	Making an object appear larger
Light microscope	Uses light
Electron microscope	Uses electrons
1 mm	1000 μm
Cell-size formula	Visible-field diameter ÷ number of cells

9. Important Questions — Topic Wise

A. Very Short Answer

Q1. What is a cell?

Ans. A cell is the basic unit of life and the basic structural and functional unit of living organisms.

Q2. Give two examples of unicellular organisms.

Ans. Bacteria and yeast.

Q3. Who observed cells in cork in 1665?

Ans. Robert Hooke.

Q4. What is magnification?

Ans. Magnification is the process of making an object appear larger than its actual size.

Q5. What is the value of 1 mm in micrometres?

Ans. $1\text{ mm} = 1000\text{ }\mu\text{m}$.

B. Short Answer

Q6. Differentiate between unicellular and multicellular organisms.

Ans. Unicellular organisms consist of only one cell, while multicellular organisms consist of many cells.

Examples: bacteria/yeast are unicellular; humans/plants are multicellular.

Q7. Why is a microscope needed to study cells?

Ans. Most cells are too small to be seen clearly by the naked eye. A microscope magnifies them and helps us observe their structure.

Q8. What did Robert Hooke observe in cork?

Ans. In 1665, Robert Hooke observed many small box-like compartments in a thin slice of cork and called them "cells".

Q9. Write the correct order of organisation in a multicellular organism.

Ans. Cell → Tissue → Organ → Organ system → Organism.

C. Numerical / Application

Q10. A visible field has a diameter of $5000\text{ }\mu\text{m}$. If 25 cells lie along the diameter, estimate the size of one cell.

Ans. Size = $5000 \div 25 = 200\text{ }\mu\text{m}$.

LAST-MINUTE REVISION

Hooke — 1665 — Cork — Cells | $1\text{ mm} = 1000\text{ }\mu\text{m}$ | Cell → Tissue → Organ → Organ system